Lab5 Exercises - Solutions

Task 1: Write a Python program to draw a line with suitable label in the x axis, y axis and a title. The values of y should be twice of x. The range of x: 1 to 30.

```
import matplotlib.pyplot as plt
X = range(1, 30)
Y = [value * 2 for value in X]
print("Values of X:")
print(*range(1,30))
print("Values of Y (twice of X):")
print(Y)
# Plot lines and/or markers to the Axes.
plt.plot(X, Y)
# Set the x axis label of the current axis.
plt.xlabel('x - axis')
# Set the y axis label of the current axis.
plt.ylabel('y - axis')
# Set a title
plt.title('Draw a line.')
# Display the figure.
plt.show()
```

Task2: Write a Python program to draw line charts of the financial data of a company between October 3, 2016 to October 7, 2016.

Sample Financial data (fdata.csv) as it appears in the .csv file:
Date,Open,High,Low,Close
10-03-16,774.25,776.065002,769.5,772.559998
10-04-16,776.030029,778.710022,772.890015,776.429993
10-05-16,779.309998,782.070007,775.650024,776.469971
10-06-16,779,780.47998,775.539978,776.859985
10-07-16,779.659973,779.659973,770.75,775.080017

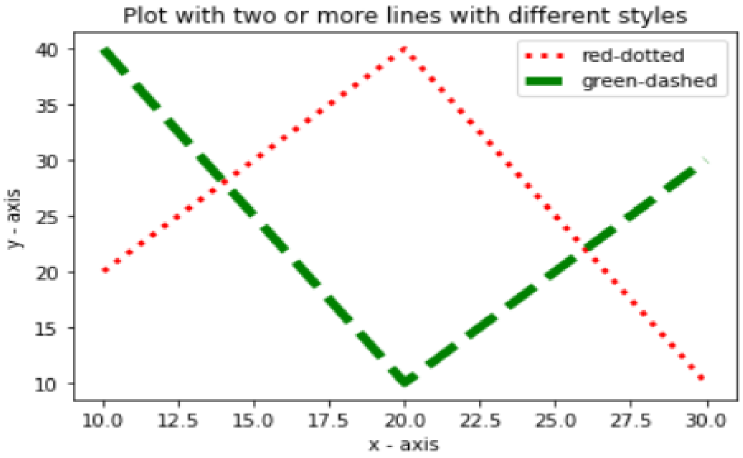
```
import matplotlib.pyplot as plt
import pandas as pd
df = pd.read_csv('fdata.csv', sep=',', parse_dates=True, index_col=0)
df.plot()
plt.show()
```

Task3: Write a Python program to plot two or more lines with legends, different widths and colours.

Sample code:

```python
import matplotlib.pyplot as plt
# line 1 points
x1 = [10,20,30,70,30]
y1 = [20,40,10,5,10]
# line 2 points
x2 = [10,20,30,70,30]
y2 = [40,10,30,5,65]
# Set the x axis label of the current axis.
plt.xlabel('x - axis')
# Set the y axis label of the current axis.
plt.ylabel('y - axis')
# Set a title
plt.title('Two or more lines with different widths and colors with suitable legends ')
# Display the figure.
plt.plot(x1,y1, color='blue', linewidth = 3,  label = 'line1-width-3')
plt.plot(x2,y2, color='red', linewidth = 5,  label = 'line2-width-5')
# show a legend on the plot
plt.legend()
plt.show()
```

Task4: Write a program that will produce the following output.




```python
import matplotlib.pyplot as plt
# line 1 points
x1 = [10,20,30]
y1 = [20,40,10]
# line 2 points
x2 = [10,20,30]
y2 = [40,10,30]
# Set the x axis label of the current axis.
plt.xlabel('x - axis')
# Set the y axis label of the current axis.
plt.ylabel('y - axis')
# Plot lines and/or markers to the Axes.
plt.plot(x1,y1, color='red', linewidth = 3,  label = 'red-dotted',linestyle='dotted')
plt.plot(x2,y2, color='green', linewidth = 5,  label = 'green-dashed', linestyle='dashed')
# Set a title
plt.title("Plot with two or more lines with different styles")
# show a legend on the plot
plt.legend()
# function to show the plot
plt.show()
```

Task5: Write a Python program to plot quantities which have an x and y position given the following arrays of x and y values:

x1 = [20, 30, 50, 60, 80]

y1 = [10, 50, 100, 180, 200]

x2 = [30, 40, 60, 70, 90]

y2 = [20, 60, 110, 200, 220]

```
import numpy as np
import pylab as pl
# Make an array of x values
x1 = [20, 30, 50, 60, 80]
# Make an array of y values for each x value
y1 = [10, 50, 100, 180, 200]
# Make an array of x values
x2 = [30, 40, 60, 70, 90]
# Make an array of y values for each x value
y2 = [20, 60, 110, 200, 220]
# set new axes limits
pl.axis([0, 100, 0, 230])
# use pylab to plot x and y as red circles
pl.plot(x1, y1,'b*', x2, y2, 'ro')
# show the plot on the screen
pl.show()
```

Task 6: Write a Python program to create multiple plots.

```
import matplotlib.pyplot as plt
fig = plt.figure()
fig.subplots_adjust(bottom=0.020, left=0.020, top = 0.900, right=0.800)

plt.subplot(2, 1, 1)
plt.xticks(()), plt.yticks(())

plt.subplot(2, 3, 4)
plt.xticks(())
plt.yticks(())

plt.subplot(2, 3, 5)
plt.xticks(())
plt.yticks(())

plt.subplot(2, 3, 6)
plt.xticks(())
plt.yticks(())

plt.show()
```

Task 7: Write a Python programming to display a bar chart of the popularity of programming Languages.

Sample data on the popularity of programming languages 2019 according to IEEE:
Programming languages: Python, Java, C, C++, R, JavaScript, C#
Popularity: 100, 96.3, 94.4, 87.5, 81.5, 79.4, 74.5

```
import matplotlib.pyplot as plt
x = ['Python', 'Java', 'C', 'C++', 'R', 'JavaScript', 'C#']
popularity = [100, 96.3, 94.4, 87.5, 81.5, 79.4, 74.5]
x_pos = [i for i, _ in enumerate(x)]
plt.bar(x_pos, popularity, color='blue')
plt.xlabel("Languages")
plt.ylabel("Popularity")
plt.title("PopularitY of Programming Language\n" + "IEEE survey 2019")
plt.xticks(x_pos, x)
# Turn on the grid
plt.minorticks_on()
plt.grid(which='major', linestyle='-', linewidth='0.5', color='red')
# Customize the minor grid
plt.grid(which='minor', linestyle=':', linewidth='0.5', color='black')
plt.show()
```

Task 8: Write a Python programming to display a horizontal bar chart of the popularity of programming Languages

Task 9: Write a Python programming to display a bar chart of the popularity of programming Languages. Use different colour for each bar.

```
import matplotlib.pyplot as plt
x = ['Python', 'Java', 'C', 'C++', 'R', 'JavaScript', 'C#']
popularity = [100, 96.3, 94.4, 87.5, 81.5, 79.4, 74.5]
x_pos = [i for i, _ in enumerate(x)]
plt.bar(x_pos, popularity, color=['red', 'black', 'green', 'blue', 'yellow', 'cyan'])

plt.xlabel("Languages")
plt.ylabel("Popularity")
plt.title("PopularitY of Programming Language\n" + "IEEE survey 2019")
plt.xticks(x_pos, x)
# Turn on the grid
plt.minorticks_on()
plt.grid(which='major', linestyle='-', linewidth='0.5', color='red')
# Customize the minor grid
plt.grid(which='minor', linestyle=':', linewidth='0.5', color='black')
plt.show()
```

Task 10: Write a Python programming to display a bar chart of the popularity of programming Languages. Attach a text label above each bar displaying its popularity (float value)

```
import matplotlib.pyplot as plt
x = ['Python', 'Java', 'C', 'C++', 'R', 'JavaScript', 'C#']
popularity = [100, 96.3, 94.4, 87.5, 81.5, 79.4, 74.5]
x_pos = [i for i, _ in enumerate(x)]

fig, ax = plt.subplots()
rects1 = ax.bar(x_pos, popularity, color=['red', 'black', 'green', 'blue', 'yellow', 'cyan'])
color=['red', 'black', 'green', 'blue', 'yellow', 'cyan']
plt.xlabel("Languages")
plt.ylabel("Popularity")
plt.title("PopularitY of Programming Language\n" + "IEEE survey 2019")
plt.xticks(x_pos, x)
# Turn on the grid
plt.minorticks_on()
plt.grid(which='major', linestyle='-', linewidth='0.5', color='red')
# Customize the minor grid
plt.grid(which='minor', linestyle=':', linewidth='0.5', color='black')

def autolabel(rects):
    """
    Attach a text label above each bar displaying its height
    """
    for rect in rects:
        height = rect.get_height()
        ax.text(rect.get_x() + rect.get_width()/2., 1.05*height,
                '%f' % float(height),
        ha='center', va='bottom')
autolabel(rects1)
plt.show()
```

Task11: Write a Python programming to create a pie chart of the popularity of programming Languages

```
import matplotlib.pyplot as plt

languages = 'Python', 'Java', 'C', 'C++', 'R', 'JavaScript', 'C#'
popularity = [100, 96.3, 94.4, 87.5, 81.5, 79.4, 74.5]
colors = ["#1f77b4", "#ff7f0e", "#2ca02c", "#d62728", "#9467bd","#8c564b","#8c364b"]

# explode 1st slice
explode = (0.1, 0, 0, 0,0,0,0)
# Plot
plt.pie(popularity, explode=explode, labels=languages, colors=colors,
autopct='%1.1f%%', shadow=True, startangle=140)

plt.axis('equal')
plt.show()
```

Task 12: Write a Python programming to create a pie chart with a title of the popularity of programming Languages. Make three wedges of the pie.

```python
import matplotlib.pyplot as plt

languages = 'Python', 'Java', 'C', 'C++', 'R', 'JavaScript', 'C#'
popularity = [100, 96.3, 94.4, 87.5, 81.5, 79.4, 74.5]
colors = ["#1f77b4", "#ff7f0e", "#2ca02c", "#d62728", "#9467bd","#8c564b","#8c364b"]

# explode 1st slice
explode = (0.1, 0, .2, 0,.1,0,0)
# Plot
plt.pie(popularity, explode=explode, labels=languages, colors=colors,
autopct='%1.1f%%', shadow=True, startangle=140)

plt.title("PopularitY of Programming Language\n" + "IEEE survey 2019")
plt.show()
```

Task 13: Create bar plot from the following DataFrame:

a b c d e

10,40,39,30,39
80,38,24,33,50
80,36,90,25,44
70,45,30,69,15
25,45,39,30,55

```python
from pandas import DataFrame
import matplotlib.pyplot as plt
import numpy as np

a=np.array([[10,40,39,30,39],[80,38,24,33,50],[80,36,90,25,44],[70,45,30,69,15],[25,45,39,30,55]])
df=DataFrame(a, columns=['a','b','c','d','e'], index=[2,4,6,8,10])

df.plot(kind='bar')
# Turn on the grid
plt.minorticks_on()
plt.grid(which='major', linestyle='-', linewidth='0.5', color='green')
plt.grid(which='minor', linestyle=':', linewidth='0.5', color='black')

plt.show()
```

Task 14: Write a Python program to create bar plots with error bars on the same figure.

Sample Date
Mean velocity: 0.2474, 0.1235, 0.1737, 0.1824
Standard deviation of velocity: 0.3314, 0.2278, 0.2836, 0.2645

```python
import numpy as np
import matplotlib.pyplot as plt
N = 5
menMeans = (54.74, 42.35, 67.37, 58.24, 30.25)
menStd = (4, 3, 4, 1, 5)
# the x locations for the groups
ind = np.arange(N)
# the width of the bars
width = 0.35
plt.bar(ind, menMeans, width, yerr=menStd, color='red')
plt.ylabel('Scores')
plt.xlabel('Velocity')
plt.title('Scores by Velocity')
# Turn on the grid
plt.minorticks_on()
plt.grid(which='major', linestyle='-', linewidth='0.5', color='green')
plt.grid(which='minor', linestyle=':', linewidth='0.5', color='black')
```

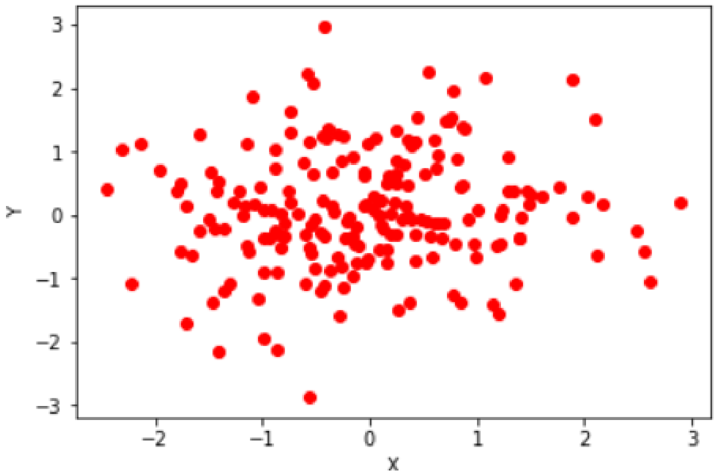
Task 15: Write a Python program to create a stacked bar plot with error bars,
Use bottom to stack the women's bars on top of the men's bars.

Sample Data:
Means (men) = (22, 30, 35, 35, 26)
Means (women) = (25, 32, 30, 35, 29)
Men Standard deviation = (4, 3, 4, 1, 5)
Women Standard deviation = (3, 5, 2, 3, 3)

```python
import numpy as np
import matplotlib.pyplot as plt

N = 5
menMeans = (22, 30, 35, 35, 26)
womenMeans = (25, 32, 30, 35, 29)
menStd = (4, 3, 4, 1, 5)
womenStd = (3, 5, 2, 3, 3)
# the x locations for the groups
ind = np.arange(N)
# the width of the bars
width = 0.35

p1 = plt.bar(ind, menMeans, width, yerr=menStd, color='red')
p2 = plt.bar(ind, womenMeans, width,
bottom=menMeans, yerr=womenStd, color='green')

plt.ylabel('Scores')
plt.xlabel('Groups')
plt.title('Scores by group\n' + 'and gender')
plt.xticks(ind, ('Group1', 'Group2', 'Group3', 'Group4', 'Group5'))
plt.yticks(np.arange(0, 81, 10))
plt.legend((p1[0], p2[0]), ('Men', 'Women'))

plt.show()
```

Task 16: Write a Python program to draw a scatter graph taking a random distribution in X and Y and plotted against each other.

```
import matplotlib.pyplot as plt
from pylab import randn
X = randn(200)
Y = randn(200)
plt.scatter(X,Y, color='r')
plt.xlabel("X")
plt.ylabel("Y")
plt.show()
```



Task 17: Write a Python program to draw a scatter plot using random distributions to generate balls of different sizes.

```
import math
import random
import matplotlib.pyplot as plt
# create random data
no_of_balls = 25
x = [random.triangular() for i in range(no_of_balls)]
y = [random.gauss(0.5, 0.25) for i in range(no_of_balls)]
colors = [random.randint(1, 4) for i in range(no_of_balls)]
areas = [math.pi * random.randint(5, 15)**2 for i in range(no_of_balls)]
# draw the plot
plt.figure()
plt.scatter(x, y, s=areas, c=colors, alpha=0.85)
plt.axis([0.0, 1.0, 0.0, 1.0])
plt.xlabel("X")
plt.ylabel("Y")
plt.show()
```

Task 19: Write a Python program to draw a scatter plot comparing two subject marks of Java and Python.

Test Data:
java_marks = [88, 92, 80, 89, 100, 80, 60, 100, 80, 34]
python_marks = [35, 79, 79, 48, 100, 88, 32, 45, 20, 30]

marks_range = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]

```
import matplotlib.pyplot as plt
import pandas as pd
java_marks = [88, 92, 80, 89, 100, 80, 60, 100, 80, 34]
python_marks = [35, 79, 79, 48, 100, 88, 32, 45, 20, 30]
marks_range = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
plt.scatter(marks_range, java_marks, label='Java marks', color='r')
plt.scatter(marks_range, python_marks, label='Python marks', color='g')
plt.title('Scatter Plot')
plt.xlabel('Marks Range')
plt.ylabel('Marks Scored')
plt.legend()
plt.show()
```